Type MS · Identify bones and bone tissues on a diagram

Answer 1 of 8 · Question 2 part a.i · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
2 (a)	(i)	1 = humerus; 2 = radius; 3 = ulna; 4 = fingers/phalanges;		4
	(ii)	circle at the elbow to include all three bones;	accept a circle around relevant bones in the fingers	1
	(iii)	• ball and socket; • gliding/sliding;	reject named examples	2
(b)		• assists/allows uptake of calcium (salts from gut); • (calcium) needed for bone growth; • hard bone/compact bone;		3

Total 10 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 1 of 8 · Question 2 part a.i · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bones and bone tissues on a diagram

Answer 2 of 8 · Question 1 part a.i · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

1	(a)	(i)	X cartilage; Y femur; Z knee cap;		1 1 1
		(ii)	**B** ligaments hold the leg bones together A is incorrect as ligaments do not attach muscles to bones C is incorrect as ligaments do not move leg bones D is incorrect as ligaments do not stop the leg bones rubbing together		1
		(iii)	hinge/synovial (joint)		1
	(b)		In the following order only oxygen; carbon dioxide;		 1 1
	(c)		biceps contracted/fatter; triceps relaxed/thinner; 		1 1

Total question 1 = 9 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 2 of 8 · Question 1 part a.i · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bones and bone tissues on a diagram

Answer 3 of 8 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

2(a)	A; (1 and 3) (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(i)	D spongy bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	B compact bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	cartilage;		**(1)**

Question Number	Answer	Additional guidance	**Mark**
2(b)(iv)	shock absorber/cushions; allows free movement of joint/prevents grinding/reduces friction;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)	• Ca^{2+}needed; • forms part of bone structure / provides strength to bone/no calcium bones are less dense/weaker; • protein required; • for muscle development/component • of muscle; • vitamin D needed; • to allow uptake of Ca^{2+};		**(6)**

Total 12 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 3 of 8 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify bones and bone tissues on a diagram

Answer 4 of 8 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

2(a)	A; (1 and 3) (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(i)	D spongy bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	B compact bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	cartilage;		**(1)**

Question Number	Answer	Additional guidance	**Mark**
2(b)(iv)	shock absorber/cushions; allows free movement of joint/prevents grinding/reduces friction;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)	• Ca^{2+}needed; • forms part of bone structure / provides strength to bone/no calcium bones are less dense/weaker; • protein required; • for muscle development/component • of muscle; • vitamin D needed; • to allow uptake of Ca^{2+};		**(6)**

Total 12 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 4 of 8 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify bones and bone tissues on a diagram

Answer 5 of 8 · Question 6 part b.i · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Question number				
6 (a)		• where two bones; • come together/meet;		2
(b)	(i)	• humerus; • ulna;		2
	(ii)	• covers end (of ball) of humerus/L; • covers depression in ulna/M;		2
(c)		• movement not smooth/bones grind/rub together/less shock absorption/increased friction; • painful;		2
(d)		any four from • attached to end of muscle/attach muscle to bone; • inelastic/do not stretch; • muscle contracts; • pulls tendon; • causing M/ulna/bone to move;		4 Total 12

Type MS · Identify bones and bone tissues on a diagram

Answer 5 of 8 · Question 6 part b.i · 2 marks

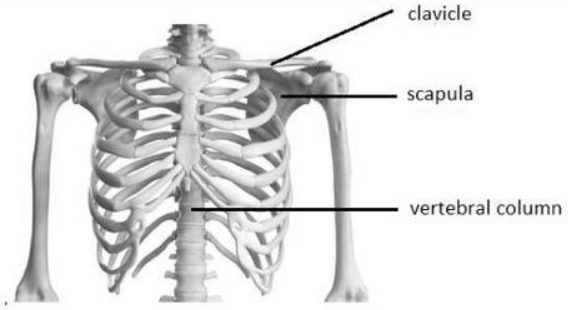
2021 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bones and bone tissues on a diagram

Answer 6 of 8 · Question 1 part a.i · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
1	(a)	(i)			3
		(ii)	• support the body; • protection (of the brain and vital organs);		2
	(b)		Three from: • decrease in bone mass/bones less dense/more spongy/porous; • bone strength weakens/brittle bones/bones break more easily/fractures occur; • height decreases/stoop/poor posture; • pain;		3

Total 8 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 6 of 8 · Question 1 part a.i · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bones and bone tissues on a diagram

Answer 7 of 8 · Question 2 part b.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
2	(a)	• two bones; • coming together/meet;		2
	(b) (i)	X = tibia; Y = femur;	reject tibula	2
	(ii)	tendons • connects bone to muscle; • transmits force/pull;		2
		cartilage • reduces friction/shock absorber/cushions; • allows smooth movement;		2
	(iii)	• ligament; • second /another muscle;		2

Total 10 marks

Type MS · Identify bones and bone tissues on a diagram

Answer 7 of 8 · Question 2 part b.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bones and bone tissues on a diagram

Answer 8 of 8 · Question 1 part a.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

1 (a) (i) long (bone) — 1

(ii)

Part	Function: reduces friction	Function: produces red blood cells	Function: provides strength
compact bone			✓
spongy bone		✓	
cartilage	✓		

3

(iii)

- provides nutrients/named nutrient/oxygen;
- removes waste/named waste

2

(b) (i) In the following order:

- tendons;
- contracts;
- relaxes;
- pulls;

4

(ii) 0.5 x 125;
62.5 (J);

Accept 63 (J)

2

Total for Question 1 = 12 marks